

Ziyu (Zoe) Meng

Frontend Engineer | React, TypeScript & Web Performance

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Authorized to work in the U.S. No sponsorship required

SUMMARY

Frontend Engineer with 4+ years building scalable web applications. Reduced B2B quote processing time by 83% through decoupled state architecture in Remix and React. Expert in performance optimization, design systems, Next.js, and Core Web Vitals tuning. Experienced in AI-integrated frontend development and agile team collaboration.

SKILLS

Frontend: React 18, TypeScript, JavaScript (ES6+), Next.js, Remix, React Router, Redux Toolkit (RTK Query), Context API, Hooks, GraphQL, HTML5, CSS3, Tailwind CSS, Responsive Design, Polaris Design System, Figma (Design-to-Code)

Performance: Core Web Vitals (LCP, CLS), Virtualization, Code Splitting, Lazy Loading, Optimistic UI

Testing: Jest, React Testing Library

Backend & Tools: Node.js, REST APIs, Shopify Admin API, Git, Prisma ORM, Drizzle ORM, AWS SES, Cloudflare Pages, Postman, CI/CD

WORK EXPERIENCE

Full Stack Web Engineer (Contract) — MTI Corporation

Mar 2022 – Present

Materials Science Equipment, Est. by MIT & UC Berkeley Researchers • Richmond, CA

- Architected a B2B quote ecosystem using Remix and Polaris Design System, reducing quote processing time by 83% through optimized server-side data fetching and parallelized API requests.
- Engineered a decoupled state management system using Redux Toolkit, enabling conflict-free simultaneous management of Retail and Wholesale carts for diverse customer cohorts.
- Built an automated document pipeline combining jsPDF, html2canvas, and AWS SES to generate client-side PDFs and trigger transactional emails, streamlining the sales closure process.
- Improved codebase reliability by implementing Jest and React Testing Library, achieving 90% test coverage for critical checkout flows and significantly reducing regression bugs.
- Partnered with cross-functional stakeholders in agile sprint cycles to translate business requirements into scalable technical solutions, establishing internal documentation standards that accelerated onboarding.

Frontend Developer (Independent Projects) — Remote

May 2020 – Feb 2022

- Built responsive marketing sites and real-time dashboards with React, Next.js (SSG/ISR), and Google Maps API, integrating location-based features for interactive user experiences.
- Optimized Core Web Vitals, reducing LCP by 25% through route-based code splitting and lazy loading strategies across multiple client projects.

PROJECTS

TagWise AI — AI-Generated SEO Tagging App

2025

- Human-in-the-Loop AI Workflow: Designed a non-destructive staging system in React 18 to manage Original vs. AI Draft states, ensuring data integrity before committing via Shopify GraphQL API.
- High-Performance Batch UI: Engineered a dashboard using Remix and virtualization to handle 100+ product catalogs with 60fps rendering and real-time SEO scoring feedback.
- Latency Optimization: Reduced perceived user latency by 40% through Optimistic UI patterns and Vite-based code splitting during live AI data stream processing.

AI Component Playground

2026

- Built an AI React component generator with Next.js 14, TypeScript, and Vercel AI SDK, streaming code generation with live Sandpack preview.
- Implemented two-phase state model for streaming/editing handoff, error resilience for incomplete code, and rate limiting to control API costs.

EDUCATION

M.A. Web Design & New Media

2018 – 2020

Academy of Art University, San Francisco

CERTIFICATIONS

Postman API Student Expert (2025) | Google Cloud Apigee Labs (2025) | SQL for Data Analysis, Coursera (2025)

Languages: English, Mandarin